开发控制台聊天客户端

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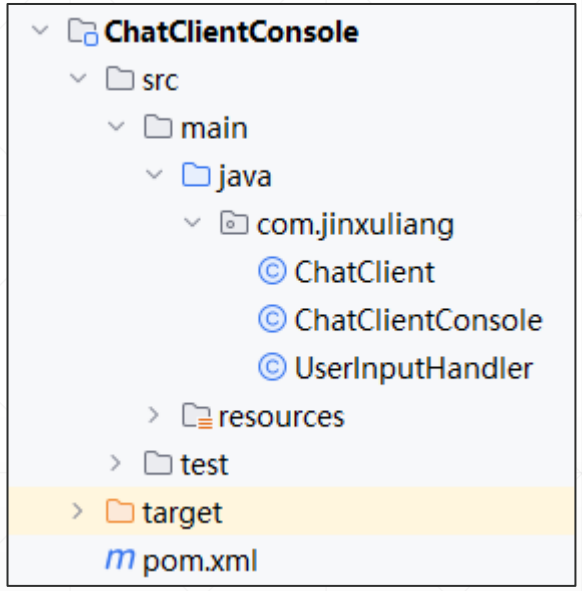
项目结构



ChatClientConsole:程序入口点

UseInputHandler:接收用户输入，发送信息

ChatClient:客户端聊天主控流程

程序入口点

```java
public class ChatClientConsole {

    public static void main(String[] args) throws IOException {
        // 等待用户输入消息
        BufferedReader consoleReader =
                new BufferedReader(new InputStreamReader(System.in));
        System.out.print("请输入您的名字(直接回车采用系统生成的名字) :");
        String userName = consoleReader.readLine();
        if (userName.isEmpty()) {
            userName = UUID.randomUUID().toString();
        }
        System.out.print("请输入服务器的IP地址:");
        String serverIP = consoleReader.readLine();
        System.out.println("本用户名:" + userName);
        ChatClient chatClient = new ChatClient(serverIP, userName);
        chatClient.start();
    }
}
```

ChatClient

```
public class ChatClient {
    7 usages
    private Socket socket;
    //接收消息
    2 usages
    private BufferedReader reader;
    //发送消息
    5 usages
    private BufferedWriter writer;

    2 usages
    private String serverIP;
    2 usages
    private String userName;

    1 usage
    public ChatClient(String serverIP, String userName) {
        this.serverIP = serverIP;
        this.userName = userName;
    }
```

```
// 检查用户是否准备退出
1 usage
public boolean readyToQuit(String msg) {
    return Constants.QUIT.equals(msg);
}

// 关闭Socket并退出
1 usage
public void close() {
    if (writer != null && !socket.isClosed()) {
        try {
            System.out.println("关闭socket");
            writer.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

ChatClient-发送消息

```
// 发送消息给服务器
1 usage
public void send(String msg) throws IOException {
    ChatMessage messageToBeSent = null;
    if (!socket.isOutputShutdown()) {
        messageToBeSent = ChatMessage.builder().userName(userName)
                .message(msg)
                .build();
        writer.write( str: ChatMessage.toJson(messageToBeSent) + "\n");
        writer.flush();

    }
}
```

ChatClient-接收消息

```
public String receive() {
    String msg = null;
    if (!socket.isInputShutdown() && !socket.isClosed()) {
        try {
            msg = reader.readLine();
        } catch (IOException e) {

        }
    }
    return msg;
}
```

ChatClient-主控流程

```
public void start() {
    try {
        // 创建socket
        socket = new Socket(serverIP, Constants.DEFAULT_SERVER_PORT);
        // 创建IO流
        reader = new BufferedReader(
                new InputStreamReader(socket.getInputStream())
        );
        writer = new BufferedWriter(
                new OutputStreamWriter(socket.getOutputStream())
        );
        // 处理用户的输入
        new Thread(new UserInputHandler( chatClient: this)).start();
        // 读取服务器转发的消息
        String msg = null;
        while ((msg = receive()) != null) {
            ChatMessage message = ChatMessage.fromJson(msg);
            System.out.println("\n" + message.getUserName() + ":" + message.getMessage());
            System.out.print(">");
        }
    } catch (IOException e) {
        e.printStackTrace();
    }
}
```